

FBISE Chemistry Class 10 Comprehensive Guess Paper 2026

Second Annual Examination

Exam date 24 September 2026

Curriculum	National Curriculum of Pakistan 2022-23
Theory Marks	65
Section A	12 MCQs 20 minutes
Sections B and C	53 marks 2 hours 40 minutes

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Complete Exam Pattern Guess Paper

FBISE Chemistry SSC II Second Annual Examination 2026

Time allowed	3 hours including Section A
Total marks	65

General Instructions

- Attempt all 12 MCQs in Section A.
- Attempt all 11 parts of Question 2. Select one alternative from each OR pair.
- Attempt Questions 3 to 6. Select one alternative in each question.
- Show formulas, substitutions, units, balanced equations, electrode labels, and observations wherever required.

Section A

Marks 12 Time 20 minutes

1. During melting at the melting point, the temperature remains constant because the supplied energy is mainly used to:
A. increase particle speed
B. overcome attractive forces
C. reduce internal energy
D. decrease particle spacing
2. At constant temperature, the volume of a fixed mass of gas is halved. According to Boyle's law, its pressure will:
A. be halved
B. double
C. remain unchanged
D. become zero
3. Which half-equation represents reduction at a cathode?
A. $\text{Cu}^{2+} + 2\text{e}^- \rightarrow \text{Cu}$
B. $2\text{Cl}^- \rightarrow \text{Cl}_2 + 2\text{e}^-$
C. $\text{Zn} \rightarrow \text{Zn}^{2+} + 2\text{e}^-$
D. $4\text{OH}^- \rightarrow \text{O}_2 + 2\text{H}_2\text{O} + 4\text{e}^-$
4. During electrolysis of dilute CuSO_4 using inert electrodes, the product at the cathode is:
A. oxygen
B. hydrogen
C. copper
D. sulfur dioxide
5. Which salt is insoluble in water according to the general solubility rules?
A. KNO_3
B. NH_4Cl
C. Na_2CO_3
D. AgCl

6. The most suitable method for preparing a pure soluble salt from an acid and an alkali is:

- A. filtration
- B. titration
- C. sublimation
- D. simple distillation

7. A horizontal line at the end of a product-concentration versus time graph shows that:

- A. the reaction is fastest
- B. the reaction has stopped
- C. activation energy is increasing
- D. the product is decomposing

8. The typical catalyst used in the Haber process is:

- A. nickel
- B. iron
- C. vanadium(V) oxide
- D. platinum

9. Which oxide is amphoteric?

- A. SO_2
- B. CaO
- C. Al_2O_3
- D. CO_2

10. Which compound immediately decolourises aqueous bromine?

- A. ethane
- B. ethene
- C. methane
- D. propane

11. Oxidation of ethanol with acidified potassium manganate(VII) produces:

- A. ethene
- B. ethanal only
- C. ethanoic acid
- D. ethyl ethanoate

12. The linkage joining amino acid residues in proteins is:

- A. ester linkage
- B. peptide linkage
- C. glycosidic linkage
- D. ionic linkage

Section B

Marks 33 Attempt all 11 parts Each part carries 3 marks

Matter (1) Explain why temperature remains constant while ice melts although heat is continuously supplied.	OR	Stoichiometry Calculate the concentration in mol/dm ³ of a solution containing 9.8 g H ₂ SO ₄ in 0.50 dm ³ . Mr H ₂ SO ₄ = 98.
Electrochemistry (2) For electrolysis of molten lead(II) chloride, state the product and observation at each electrode and write both ionic half-equations.	OR	Electrochemistry Given E values Zn ²⁺ /Zn = -0.76 V and Cu ²⁺ /Cu = +0.34 V, identify the more reactive metal and justify the order using the reduction potentials.
Electrochemistry (3) State the reactants, overall equation, and one advantage of a hydrogen-oxygen fuel cell.	OR	Salts Why are salts normally crystalline solids with high melting points, yet conduct electricity only when molten or aqueous?
Nitrogen and Sulfur (4) Write the Haber process equation and state the sources of nitrogen and hydrogen and the typical operating conditions.	OR	Nitrogen and Sulfur Write the Contact process equation and state the sources of reactants and the typical conditions for converting SO ₂ to SO ₃ .
Reaction Kinetics (5) A product-concentration graph rises steeply at first and then becomes horizontal. Explain what each part shows about reaction rate.	OR	Reaction Kinetics Explain two ways in which control of reaction rate is important during harvesting, storage, or transport of food.
Oxides (6) Classify SO ₂ , CaO, and ZnO as acidic, basic, or amphoteric. Support each classification with one suitable equation.	OR	Metals Describe the reactions of a suitable metal with cold water, steam, and dilute acid. Write balanced equations.
Organic Chemistry (7) Draw structural or displayed formulae for but-2-ene, propan-2-ol, and ethanoic acid.	OR	Organic Chemistry Name and draw the ester formed from methanol and propanoic acid, and write its formation equation.
Hydrocarbons (8) Describe the bromine-water and acidified or alkaline KMnO ₄ tests used to distinguish ethene from ethane, including observations.	OR	Hydrocarbons Prepare ethene by two elimination reactions: dehydration of ethanol and dehydrohalogenation of bromoethane. Give equations and conditions.

Hydrocarbons (9) What is fractional distillation of petroleum? Name two fractions and give one use of each.	OR	Energy Name the three principal fossil fuels, identify the main constituent of natural gas, and explain why petroleum is a mixture.
Hydroxy Compounds (10) Compare fermentation and catalytic hydration of ethene for ethanol manufacture by giving one condition, one advantage, and one disadvantage of each.	OR	Hydroxy Compounds Write the complete combustion equation of ethanol and state two advantages and one disadvantage of alcohol fuels compared with fossil fuels.
Carboxylic Compounds (11) Write equations for ethanoic acid reacting with magnesium, sodium hydroxide, and sodium carbonate. Name the salt in each case.	OR	Polymers Differentiate addition and condensation polymerisation, and identify the linkage formed when a dicarboxylic acid reacts with a diamine.

Section C

Marks 20 Attempt all questions Each question carries 5 marks

Question 3 Marks 5

A heating curve for ice is drawn from below 0 °C to steam above 100 °C.

- (a) Sketch and label the curve.
- (b) Explain the sloping and horizontal regions using kinetic particle theory and internal energy.
- (c) Distinguish boiling from evaporation.

OR

Describe electrolysis of concentrated aqueous sodium chloride using inert electrodes.

- (a) Draw and label the cell.
- (b) State ion movement, products, and observations.
- (c) Write both ionic half-equations and indicate electron flow.

Question 4 Marks 5

A compound contains 40.0 percent carbon, 6.67 percent hydrogen, and 53.33 percent oxygen. Its molar mass is 180 g/mol.

- (a) Determine its empirical formula.
 - (b) Determine its molecular formula.
- Show all steps.

OR

Describe the general chemical properties of metals through their reactions with oxygen, cold water, steam, and dilute acids. Give one balanced equation for each applicable reaction and relate the observations to metal reactivity.

Question 5 Marks 5

Explain fractional distillation of petroleum. Describe how chain length, boiling point, volatility, and viscosity change from the bottom to the top of the column, and state uses of four fractions.

OR

Explain the chemistry of alkenes. Include cracking, the reason for cracking, preparation of ethene from ethanol and a halogenoalkane, tests for unsaturation, and two addition reactions with equations.

Question 6 Marks 5

Describe both methods of manufacturing ethanol with equations and conditions. Compare their advantages and disadvantages, then explain complete combustion and two industrial or daily-life uses of alcohols.

OR

Explain condensation polymerisation between a dicarboxylic acid and a diamine. Show the general repeating linkage, compare it with addition polymerisation, and discuss three environmental problems caused by plastic disposal.

Answer Key for Section A

1. B	2. B	3. A	4. C
5. D	6. B	7. B	8. B
9. C	10. B	11. C	12. B

High Yield Revision Priority

Level	Preparation focus
Priority A	Electrochemistry; stoichiometric calculations; hydrocarbons and petroleum; metals and oxides; ethanol and carboxylic acids; polymers
Priority B	Matter and gas laws; salts; Haber and Contact processes; reaction-rate graphs; proteins and vitamins; fossil fuels
Answer standard	Definitions plus reasons; balanced equations; labelled diagrams; observations; complete numerical working with units

Extra High Yield MCQ Bank

Prepare these in addition to the 12 MCQs in the exam-pattern set

1. The direct change of a gas into a solid is called:

- A. freezing
- B. deposition
- C. condensation
- D. sublimation

2. The rate of diffusion of a gas generally increases when:

- A. molecular mass increases
- B. temperature decreases
- C. temperature increases
- D. pressure becomes zero

3. At constant pressure, the volume of a gas is directly proportional to its absolute temperature. This is:

- A. Boyle's law
- B. Charles's law
- C. Avogadro's law
- D. Dalton's law

4. Increasing external pressure generally causes the boiling point of a liquid to:

- A. decrease
- B. increase
- C. remain fixed
- D. become 0 C

5. The unit mol/dm³ represents:

- A. mass concentration
- B. molar concentration
- C. molar mass
- D. percentage yield

6. The empirical formula is:

- A. the actual number of atoms
- B. the simplest whole-number ratio of atoms
- C. always equal to molecular formula
- D. the molar mass

7. The number of dm³ in 500 cm³ is:

- A. 0.050
- B. 0.50
- C. 5.0
- D. 500000

8. In an electrolytic cell, oxidation occurs at the:

- A. cathode
- B. electrolyte only
- C. anode
- D. salt bridge

9. Electrons in the external circuit of an electrolytic cell move:

- A. through the electrolyte
- B. from the power source toward the cathode and away from the anode
- C. only from cathode to anode
- D. with the cations

10. Electrolysis of molten lead(II) chloride produces chlorine at the:

- A. cathode
- B. anode
- C. salt bridge
- D. power supply

11. Using copper electrodes in CuSO_4 solution, the anode:

- A. gains mass
- B. dissolves
- C. produces hydrogen
- D. remains completely unchanged

12. The only chemical product of a hydrogen-oxygen fuel cell is:

- A. carbon dioxide
- B. ammonia
- C. water
- D. methane

13. In a Daniel cell, the salt bridge mainly:

- A. supplies electrons
- B. maintains electrical neutrality
- C. increases temperature
- D. acts as a catalyst

14. A metal with a more negative standard reduction potential is generally:

- A. less easily oxidised
- B. more reactive
- C. a stronger oxidising agent
- D. always a non-metal

15. A steeper slope on a product-concentration versus time graph indicates:

- A. a faster rate
- B. a slower rate
- C. no reaction
- D. constant activation energy only

16. A rapid rise in temperature during a reaction is a physical sign of:

- A. a higher reaction rate
- B. a lower molarity
- C. a smaller atomic mass
- D. electrolysis

17. All nitrate salts are generally:

- A. insoluble
- B. soluble
- C. gases
- D. amphoteric

18. Which carbonate is soluble?

- | | |
|----------------------------|--------------------|
| A. CaCO_3 | B. CuCO_3 |
| C. K_2CO_3 | D. PbCO_3 |

19. Excess insoluble base is removed from a salt solution by:

- | | |
|-----------------|----------------|
| A. filtration | B. titration |
| C. distillation | D. sublimation |

20. PAN is a component of:

- | | |
|------------------------|-----------------------|
| A. acid-base titration | B. photochemical smog |
| C. hard water | D. natural gas |

21. The source of nitrogen in the Haber process is:

- | | |
|------------|-----------|
| A. methane | B. air |
| C. water | D. sulfur |

22. The catalyst used to convert SO_2 to SO_3 in the Contact process is:

- | | |
|---------------------------|-------|
| A. Fe | B. Ni |
| C. V_2O_5 | D. Cu |

23. A basic oxide among the following is:

- | | |
|------------------|----------------------------|
| A. CO_2 | B. SO_2 |
| C. CuO | D. Al_2O_3 |

24. An amphoteric oxide reacts with:

- | | |
|-------------------------|---------------|
| A. acids only | B. bases only |
| C. both acids and bases | D. neither |

25. The suffix -ol identifies a compound as an:

- | | |
|-----------|--------------------|
| A. alkene | B. alcohol |
| C. ester | D. carboxylic acid |

26. Cracking converts large alkane molecules mainly into:

- | | |
|--------------------------------|----------------|
| A. smaller alkanes and alkenes | B. salts |
| C. metals | D. amino acids |

27. Hydrogenation of ethene uses:

- | | |
|----------------------------|------------------------|
| A. steam and acid catalyst | B. hydrogen and nickel |
| C. bromine water | D. oxygen and iron |

28. Ethyne is useful in welding because its combustion flame is:

- A. very cold
- B. very hot
- C. colourless only
- D. non-reactive

29. From bottom to top of a fractionating column, petroleum fractions become:

- A. less volatile
- B. more viscous
- C. lower boiling
- D. longer chained

30. The petroleum fraction used for road surfacing is:

- A. naphtha
- B. kerosene
- C. bitumen
- D. refinery gas

31. Fermentation of glucose requires yeast and:

- A. oxygen-rich conditions
- B. absence of oxygen
- C. 200 atm pressure
- D. V₂O₅

32. Catalytic hydration of ethene uses approximately:

- A. 300 C and an acid catalyst
- B. 25 C and yeast
- C. 450 C and iron
- D. room temperature only

33. When a carboxylic acid reacts with a carbonate, the gas evolved is:

- A. H₂
- B. O₂
- C. CO₂
- D. Cl₂

34. An ester is formed from:

- A. an alcohol and a carboxylic acid
- B. an alkane and oxygen
- C. a metal and acid
- D. two amino acids only

35. Condensation polymerisation usually produces:

- A. no by-product
- B. a small molecule such as water
- C. only oxygen
- D. a metal

36. A polyamide is formed from a dicarboxylic acid and a:

- A. diamine
- B. diol
- C. alkene
- D. monohydric alcohol

37. Burning plastics may release:

- A. only water
- B. toxic gases
- C. only oxygen
- D. nitrogen fertilizer

38. Proteins are natural polymers of:

- A. fatty acids
- B. amino acids
- C. glucose only
- D. nucleotides

39. Vitamin D is especially important for:

- A. calcium absorption and bones
- B. forming petroleum
- C. electroplating
- D. cracking alkanes

40. The main constituent of natural gas is:

- A. ethene
- B. methane
- C. ethanol
- D. carbon monoxide

Answer Key for Extra MCQ Bank

1. B	2. C	3. B	4. B	5. B
6. B	7. B	8. C	9. B	10. B
11. B	12. C	13. B	14. B	15. A
16. A	17. B	18. C	19. A	20. B
21. B	22. C	23. C	24. C	25. B
26. A	27. B	28. B	29. C	30. C
31. B	32. A	33. C	34. A	35. B
36. A	37. B	38. B	39. A	40. B

Extra High Yield Short Questions

Priority A questions should be prepared first

Priority A Must Prepare

1. Interpret a heating or cooling curve and explain why temperature stays constant during a change of state.
2. Differentiate evaporation and boiling, and explain the effect of external pressure on each.
3. Explain diffusion using kinetic particle theory and discuss the effects of molecular mass and temperature, including medicine delivery in the body.
4. Convert concentration between g/dm^3 and mol/dm^3 and solve a numerical involving solution volume in cm^3 and dm^3 .
5. Determine empirical and molecular formula from percentage-composition data and molar mass.
6. Describe electrolysis of concentrated aqueous NaCl using inert electrodes with products, observations, and half-equations.
7. Describe electrolysis of dilute H_2SO_4 using inert electrodes with products, observations, and half-equations.
8. Compare electrolysis of dilute CuSO_4 using inert electrodes and copper electrodes.
9. Explain a hydrogen-oxygen fuel cell and compare it with a petrol engine.
10. Describe electroplating, identify anode, cathode, and electrolyte, and give three applications.
11. Sketch a labelled Daniel cell and use supplied standard potentials to arrange metals in reactivity order.
12. Interpret mass-time, gas-volume-time, temperature-time, and concentration-time reaction-rate graphs.
13. Explain how reaction-rate control supports harvesting, storage, ripening, and transport in the food industry.
14. Explain ionic lattice structure, high melting point, state at STP, and electrical conductivity of salts.
15. State the general solubility rules and select suitable methods to prepare salts from acids and alkalis, metals, bases, or carbonates.
16. Explain how NO and NO_2 react with unburned hydrocarbons to form PAN in photochemical smog.
17. Write the Haber process equation, sources, conditions, catalyst, and purpose.
18. Write the Contact process equation, sources, conditions, catalyst, and purpose.
19. Classify SO_2 , CO_2 , CuO , CaO , Al_2O_3 , and ZnO and support classifications with equations.
20. Describe reactions of metals with oxygen, cold water, steam, and dilute acids using balanced equations.
21. Name or draw unbranched alkanes, alkenes, alcohols, carboxylic acids, and esters from formulae or names.
22. Explain cracking, why it is needed, and write a valid example equation.
23. Distinguish saturated and unsaturated hydrocarbons using bromine water and KMnO_4 , with observations and equations.
24. Write equations for alkene addition with bromine, hydrogen, and steam, and prepare an alkene by two elimination methods.
25. Describe fractional distillation and the trend in chain length, volatility, boiling point, and viscosity through the column.

26. Compare fermentation and catalytic hydration of ethene for manufacturing ethanol.
27. Describe oxidation and combustion of ethanol and discuss uses of alcohols as fuels, solvents, disinfectants, and industrial materials.
28. Describe reactions of carboxylic acids with metals, bases, and carbonates and explain esterification.
29. Compare addition and condensation polymerisation and deduce the linkage produced from a dicarboxylic acid with a diamine or a diol.
30. Explain why proteins are natural polyamides, draw the general amino-acid structure, show peptide linkage formation, and state sources and importance of major vitamins.

Priority B Additional Variants

1. Explain Boyle's, Charles's, and Avogadro's laws qualitatively using kinetic particle theory.
2. Explain why lighter gases diffuse faster and predict the effect of heating on diffusion rate.
3. Predict electrolysis products for a dilute and a concentrated halide solution.
4. Explain charge transfer in the external circuit, ion movement in electrolyte, and electron transfer at electrodes.
5. Prepare a named soluble salt by titration and by reaction with excess insoluble solid.
6. Arrange metals from experimental observations with water and dilute acid.
7. Explain the uses of ethyne in welding and artificial fruit ripening, including a safety concern.
8. List eight petroleum fractions and one use of each.
9. Explain formation of ethanoic acid by chemical oxidation and by bacterial oxidation in vinegar production.
10. Discuss industrial and daily-life uses of carboxylic acids and esters.
11. Discuss landfill accumulation, ocean pollution, and toxic gases from burning plastics.
12. Outline why polymers are important in textiles and relate properties to uses.
13. State the major fossil fuels, explain why petroleum is a hydrocarbon mixture, and identify methane as the main constituent of natural gas.
14. Explain the functions, sources, and deficiency effects of vitamins A, C, D, and K.

Extra High Yield Long Questions

Use equations, diagrams, observations, comparisons, and reasons

1. Matter. Explain all six changes of state using kinetic particle theory and internal energy. Interpret heating and cooling curves and distinguish boiling from evaporation.
2. Gases and diffusion. Explain Boyle's, Charles's, and Avogadro's laws qualitatively. Discuss how temperature and molecular mass affect diffusion and why diffusion matters in medicine delivery.
3. Stoichiometry. Solve a multi-step problem involving reacting masses, gas volume at RTP, solution concentration, and empirical and molecular formula. Show units and balanced equations.
4. Electrolysis. Compare the electrolysis of molten lead(II) chloride, concentrated aqueous NaCl, dilute H₂SO₄, and dilute CuSO₄ using appropriate electrodes. Include products, observations, and half-equations.
5. Electrochemical applications. Explain the hydrogen-oxygen fuel cell, electroplating, and the Daniel cell. Include diagrams, equations, electron direction, and advantages or applications.
6. Salts. Explain ionic lattice properties, conductivity, solubility rules, and preparation and purification of soluble salts by four acid-reaction methods.
7. Reaction kinetics. Interpret experimental rate graphs and explain how mass, temperature, and gas formation can indicate rate. Discuss the importance of rate control in the food industry.
8. Industrial inorganic chemistry. Describe the Haber and Contact processes with equations, reactant sources, conditions, and catalysts. Explain PAN formation from nitrogen oxides and unburned hydrocarbons.
9. Oxides and metals. Classify selected oxides as acidic, basic, or amphoteric with equations, then describe and compare metal reactions with oxygen, cold water, steam, and dilute acids.
10. Alkenes. Explain cracking and elimination methods of manufacture; tests for unsaturation; and addition with bromine, hydrogen, and steam, using displayed formulae and equations.
11. Petroleum. Describe fractional distillation, property trends in the column, names and uses of fractions, and reasons for cracking large hydrocarbons.
12. Alcohols. Compare fermentation and catalytic hydration of ethene, explain alcohol combustion, and evaluate alcohols as fuels and as materials in pharmaceuticals, cosmetics, solvents, and disinfectants.
13. Carboxylic acids and esters. Explain ethanoic acid formation, reactions with metals, bases, and carbonates, esterification, and key industrial and daily-life applications.
14. Polymers. Compare addition and condensation polymerisation; explain polyamide and polyester formation from monomers; discuss textile uses and environmental challenges of plastics.
15. Biochemistry and energy. Explain protein formation and peptide linkages, discuss vitamins and health, and summarize coal, natural gas, petroleum, methane, and the hydrocarbon nature of petroleum.

Numerical Practice Set

1. A solution contains 5.85 g NaCl in 250 cm³. Calculate concentration in g/dm³ and mol/dm³. Mr NaCl = 58.5.
2. How many grams of NaOH are needed to prepare 500 cm³ of 0.20 mol/dm³ solution? Mr NaOH = 40.
3. A compound contains 54.55 percent C, 9.09 percent H, and 36.36 percent O. Its molar mass is 88 g/mol. Determine empirical and molecular formula.
4. A compound has empirical formula CH₂O and molar mass 180 g/mol. Determine its molecular formula.
5. Using $2\text{CO} + \text{O}_2 \rightarrow 2\text{CO}_2$, calculate the mass and RTP volume of CO₂ formed from 28 g CO. Mr CO = 28; molar gas volume = 24 dm³/mol.
6. Calcium carbonate reacts as $\text{CaCO}_3 + 2\text{HCl} \rightarrow \text{CaCl}_2 + \text{CO}_2 + \text{H}_2\text{O}$. Calculate the mass of CaCl₂ formed from 25.0 g CaCO₃ when HCl is in excess. Mr CaCO₃ = 100; Mr CaCl₂ = 111.
7. A gas occupies 300 cm³ at 300 K. Predict its volume at 360 K at constant pressure.
8. A gas occupies 500 cm³ at 100 kPa. Predict its volume at 250 kPa at constant temperature.

Essential Diagrams to Practise

1. Heating and cooling curves with labelled phase regions.
2. Electrolytic cell showing power supply, anode, cathode, electrolyte, ion movement, and electron direction.
3. Electrolysis of concentrated brine and dilute copper(II) sulfate.
4. Daniel cell with both half-cells, salt bridge, voltmeter, and electron flow.
5. Fractionating column with fraction order and property trends.
6. General amino acid, dipeptide or tripeptide, and peptide linkage.
7. General formation of a polyamide and a polyester, showing the repeating linkage.

Scope Control for Theory Preparation

Do not treat the following as theory priorities for this examination because the supplied framework marks them formative, PBA-only, or not for annual theory: History of Chemistry; percentage yield, percentage purity, and percentage composition calculations; limiting-reactant calculations; collision theory and general factor explanations; acid-rain chemistry from NO_x; deduction of an addition-polymer repeat unit where the framework remarks exclude it; detailed nylon and PET structures and PET depolymerisation; detailed structures, sources, and uses of carbohydrates and lipids; nucleic-acid importance; biochemical testing, genetic engineering, gene therapy, and cloning; laboratory hazards and PBA-only tasks.

Borderline point. Gas-volume calculations at RTP remain useful because the summative stoichiometry SLO includes gas-volume relationships, even though a separate molar-gas-volume SLO is labelled formative. Prepare them after the core Priority A questions.

Final Student Checklist

- [] I can write every required balanced symbol and ionic half-equation without help.
- [] I can draw and label the electrolysis, Daniel-cell, fractionation, curve, protein, and polymer diagrams.
- [] I show the formula, substitution, working, final answer, and unit in every numerical.
- [] I prepare both alternatives in every OR pair and do not rely on selective study.
- [] I use precise observations such as colour change, gas formation, electrode deposit, and mass change.